

ENGR 102 Summer Bridge Day 4 Student Quiz

Comparisons, Exact Boundaries, and Ordered Branches

20 points • approximately 20-25 minutes

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Boolean-operator convention. Python operators appear as **and**, **or**, and **not**. Their lowercase spelling is required in code.

Show intermediate state for trace credit. Program credit requires one coherent solution. Unless a prompt says otherwise, give exact Python 3 output.

Question 1. Evaluate exact-boundary comparisons.

5 points

```
low = 10
high = 15
value = 15
within = low <= value <= high
above = value > high
print(within, above)
```

Evaluate every comparison and explain which operator controls the upper boundary.

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Question 2. Trace the first matching ordered branch.

5 points

```
reading = 25

if reading > 25:
    status = "high"
elif reading >= 20:
    status = "normal"
else:
    status = "low"

print(status)
```

Name the first matching branch, give exact output, and state what happens at 25 and 25.1.

Question 3. Program construction

10 points

Write one complete Python program that satisfies every requirement.

- Read temperature as a float.
- Print invalid when temperature is below -40 or above 80.
- For valid input, classify cold below 15, warm from 15 through 30 inclusive, and hot above 30.
- Print exactly Status: followed by the classification using one if/elif/elif/else chain.

[illegible]

Test evidence / expected result